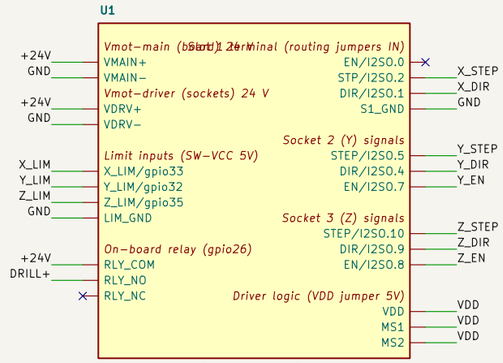
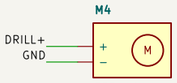


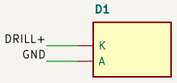
24 V 6 A brick + barrel-to-screw adapter



PiBot V4.96 Pro (ESP32, FluidNC)



555 drill motor (24 V)



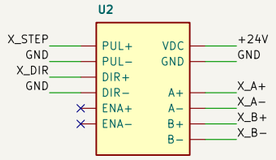
1N5822 flyback (recommended)

X index (nail 0)

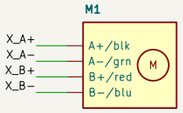
Y home (tip outside)

Z home (drill up)

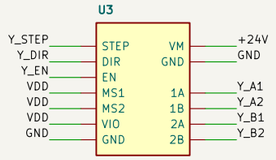
NOTES
1. Pins follow the PiBot V4.96 Pro factory YAML (wiki.pibot.com, checked 2026-11-04) and ../fluidnc-config.yaml. Steppers are on the I2S shift register (I2SO.n). gpio2 is the probe INPUT.
2. Board jumpers: VDD = 5V, SW-VCC = 5V, GPIO26 = Relay Output, USB-5V removed, SPI removed, MS1 + MS2 fitted (1/16), RST/SLP off. Slot 1 routing jumpers IN and its socket EMPTY; slots 2-6 routing jumpers OUT. A driver in a jumpered slot destroys the driver and the board.
3. Power: 24 V to BOTH Vmot-main (board) and Vmot-driver (socket drivers), plus the DM542T VDC. Recommended: inline 5-6 A fuse after J1. Never unplug a motor with power on.
4. DM542T common cathode: STP to PUL+, DIR to DIR+, slot GND to PUL- and DIR-, ENA open. DIP SW1-3 = OFF ON OFF (2.36 A RMS), SW4 OFF, SW5-8 = ON ON OFF ON (3200 pulses/rev).
5. TMC2209 standalone, 1/16: MS1 = MS2 = HIGH via the shared MS caps. Vref 0.9-1.0 V, max 1.4 V.
6. Endstops COM to GND, NC to the input: closed at rest, opens when hit or if a wire breaks; config pins have no :low. Input LEDs are lit at rest and go out when a switch is pressed.
7. Drill through the on-board relay: COM from +24 V, NO to DRILL+. D1 across the drill, stripe to +.
8. Coil pairs: black/green = A (1A/1B), red/blue = B (2A/2B). Swap one pair to reverse an axis.



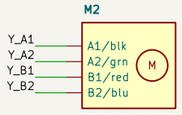
DM542T, DIP 1-8: OFF ON OFF OFF ON ON OFF ON



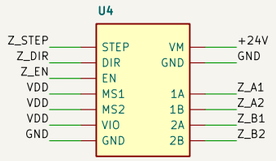
23HS45-4204S platter (X)



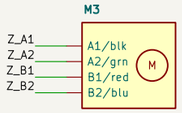
TMC2209 in socket 2 (Y), Vref 0.9-1.0 V



17HE08-1004S threader (Y)



TMC2209 in socket 3 (Z), Vref 0.9-1.0 V



17HE08-1004S drill Z

Pins mirror machine/fluidnc-config.yaml
Generated by machine/wiring/make_schematic.py
crobison/string-art

Sheet: /
File: wiring.kicad_sch

Title: String art machine wiring

Size: A3	Date: 2026-11-04	**Rev: 2**
KiCad E.D.A. 10.0.8		Id: 1/1